

Debarghya Roy (he/him)

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EDUCATION

University of Illinois at Urbana Champaign (UIUC)

Aug. 2025 – Expected May 2029

Bachelor of Science in Electrical Engineering

Champaign, IL

- Cumulative GPA: **4.00 / 4.00**, Honors: James Scholar, Dean's List
- Relevant Coursework: ECE 210 – Analog Signal Processing, ECE 145 – First-Year Design Laboratory, PHYS 214 – Quantum Physics, PHYS 213 – Thermal Physics, MATH 257 – Linear Algebra with Computational Applications, MATH 285 – Intro to Differential Equations

Harvard University

Mar. 2024

CS50 Certification

Remote

- Successfully completed the **Introduction to Python** and **Introduction to Artificial Intelligence** courses

EXPERIENCE

Eco Illini Supermileage

Aug. 2025 – Present

Electrical Team Member

Champaign, IL

- Participated in the **2026 Shell Eco Marathon Competition** and achieved an efficiency of **186.47 km/kWh**
- Engineered a **97%-efficient** buck converter (32-42V to 12V @ 5A) as part of an electric vehicle's Power Distribution Unit
- Implemented **isolated power architecture** and **kill-switch** safety mechanisms for **reliable** subsystem power delivery
- Enabled **real-time energy monitoring** via INA228 and STM32G0B1 over CAN FD

Illinois MicroTech

Oct. 2025 – Present

Optics and Electronics Team Member

Champaign, IL

- Member of the CMU based Hackerfab project, a student led initiative created to **make nanofabrication more accessible**
- Strategized and **implemented solutions** to resolve key operational challenges with the **Lithography Stepper**
- Investigated and mitigated UV lithography imaging artifacts and experimentally determined optical resolution and feature size accuracy down to **0.2um/pixel**

Science Reach, Adlai E. Stevenson High School

Aug. 2023 – May 2025

Researcher

Lincolnshire, IL

- Authored [“Comparative Analysis of CNNs for Classifying Alzheimer's Disease”](#)
- Achieved **95% accuracy** using VGG16 model with the Convolutional Block Attention Mechanism
- Improved early detection of Alzheimer's for more than **55 million patients with dementia**
- Earned the Jay and Jeremy Hoffman Memorial Award and **Gold Honors** at Illinois Junior Academy of Sciences State Science Fair

Parabole.ai

May 2024 – Aug. 2024

Intern

Remote

- **Led a team of four interns** to migrate the company's website from Flask to React, **improving site responsiveness**
- Integrated the company's TRAIN platform API to **generate causal inference graphs**, enhancing product usability for **100+ users**
- Collaborated with project leaders to **deliver production-ready features** in 2 months

SERVICE & PROJECTS

ECE 145: First Year Design Laboratory

Aug. 2025 – Dec. 2025

Honors

Champaign, IL

- Created a **remote-controlled, multi-colored, automatic** lighting system that adjusts to ambient light using **analog circuitry**
- Maintained a **detailed journal** and project proposal to document the whole design and development process

The Parris Foundation: S.T.E.M.ulating Minds

Aug. 2022 – May 2025

Volunteer

Remote

- Tutored underprivileged 6-12th grade students in **Computer Science, Python, and Mathematics** on a weekly basis
- Fostered academic growth by providing **personalized guidance** and support to students
- Recognized with **4 awards** for Volunteer and Tutor dedication

TECHNICAL SKILLS

- **Languages & Libraries/Frameworks:** C, ReactJS, Java, Python, TensorFlow, Scikit Learn, Pandas, NumPy, Unix/Linux Shell
- **Tools:** KiCad, Raspberry Pi, STM32CubeIDE, Oscilloscopes, Function Generators, Git